

Article

MA-HRL: Multi-Agent Hierarchical Reinforcement Learning for Medical Diagnostic Dialogue Systems

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Abstract

Task-oriented medical dialogue systems face two fundamental challenges: the explosion of state-action space caused by numerous diseases and symptoms and the sparsity of informative signals during interactive diagnosis. These issues significantly hinder the accuracy and efficiency of automated clinical reasoning. To address these problems, we propose MA-HRL, a multi-agent hierarchical reinforcement learning framework that decomposes the diagnostic task into specialized agents. A high-level controller coordinates symptom inquiry via multiple worker agents, each targeting a specific disease group, while a two-tier disease classifier refines diagnostic decisions through hierarchical probability reasoning. To combat sparse rewards, we design an information entropy-based reward function that encourages agents to acquire maximally informative symptoms. Additionally, medical knowledge graphs are integrated to guide decision-making and improve dialogue coherence. Experiments on the SymCat-derived SD dataset demonstrate that MA-HRL achieves substantial improvements over state-of-the-art baselines, including +7.2% diagnosis accuracy, +0.91% symptom hit rate, and +15.94% symptom recognition rate. Ablation studies further verify the effectiveness of each module. This work highlights the potential of hierarchical, knowledge-aware multi-agent systems for interpretable and scalable medical diagnosis.

Keywords: multi-agent reinforcement learning; medical diagnostic dialogue systems; information entropy reward; knowledge graph



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1. Introduction

In recent years, the field of deep learning, particularly in reinforcement learning (RL), has witnessed significant advancements, garnering attention for its capability to extract intricate patterns from massive datasets [1–5]. Reinforcement learning algorithms are primarily utilized to address the challenge of determining how intelligent agents (referred to as “Agents”) should act within their surrounding environments to maximize cumulative rewards [6–9]. Similar to humans, RL agents use interactive learning to successfully obtain satisfactory decision strategies [10–13]. Within this context, task-oriented dialogue systems based on deep reinforcement learning (DRL) have also found successful applications in the medical domain [14–16]. RL agents, through interactive conversations with patients, acquire critical pathological features from patients and subsequently assist healthcare

professionals in diagnosing illnesses. This has, to some extent, alleviated the issue of constrained medical resources. However, the sheer volume of diseases and symptoms has introduced challenges related to the dimensions of action and state spaces, leading to sparse reward problems, consequently impacting the accuracy of disease discrimination [17].

To address this issue, the current approach involves integrating a knowledge graph with the dialogue system to prune the action space, to some extent, enhancing the diagnostic accuracy of intelligent diagnostic dialogue systems [18,19]. However, existing intelligent diagnostic dialogue systems heavily rely on data-driven learning and often fail to truly comprehend the relationship between symptoms and diseases. Consequently, most research on disease diagnosis dialogue systems is conducted without medical knowledge or relies heavily on statistical features (i.e., conditional probabilities from symptoms to diseases) [20–22]. Decision-making is based on the dialogue state, without effectively resolving the dimensionality challenge and sparse reward problem.

To address the aforementioned issues, this paper presents an intelligent diagnostic dialogue algorithm for disease diagnosis based on information entropy-guided hierarchical reinforcement learning, drawing inspiration from the human approach to dealing with complex problems [23–26]. Specifically, we restructure the problem based on clinical conversations, which primarily involves modeling the state space and reward function to accurately reflect the patient case information within the dialogue system.

Furthermore, we introduce the concept of information entropy by incorporating rewards that measure the information difference between consecutive states, thereby incentivizing the intelligent agent to actively seek more symptoms from the user [27–30]. This paper introduces an information entropy-based hierarchical reinforcement learning framework, which effectively mitigates the problem of state sparsity by integrating a hierarchical structure into the disease classifier. Lastly, this study incorporates prior medical knowledge graphs to further enhance the task completion rate of the dialogue system while overcoming some of the limitations of purely data-driven learning.

In summary, this paper makes the following contributions:

1. We propose an intelligent diagnostic dialogue management algorithm based on a hierarchical reinforcement learning framework. The hierarchical architecture alleviates the dimensionality disaster problem caused by the large number of diseases and symptoms in the action and state spaces.
2. We address the issue of sparsity in the disease classifier. We introduce the concept of information entropy and propose an information entropy-based reward function.
3. Furthermore, considering the higher requirement for coherence in medical dialogue, we enhance the diagnostic accuracy of the dialogue system by incorporating a medical knowledge graph.

We evaluate the MA-HRL model using the SD dataset [31]. The experimental results demonstrate that our model outperforms state-of-the-art methods in various evaluation metrics, except for the number of dialogue turns. Specifically, we achieve improvements of 7.2% in diagnosis accuracy, 0.91% in symptom hit rate, and 15.94% in symptom recognition rate.

2. Related Works

Reinforcement-learning-based dialogue systems have achieved significant progress in various domains in the past decade. In 2018, Wei et al. [21] first proposed combining dialogue systems with disease information collection and automated diagnosis tasks. They introduced a reinforcement-learning-based dialogue framework and annotated the first dataset for training intelligent diagnostic dialogue systems. However, this dataset had limited coverage of diseases and symptoms. In their dialogue management module, they

only used a single DQN model, where the action space consists of a combination of diseases and symptoms. The size of the action space was the sum of the number of diseases and symptoms, which would be very large in real-world scenarios, leading to the dimensionality disaster problem.

To address this issue, Tang et al. [32,33] segmented diseases into several groups and trained corresponding strategies for each group, reducing the action space size for each model. They employed fixed rules to determine which strategy to interact with patients. Building upon this, Kao et al. [20] improved upon this approach by no longer relying on fixed rules to select sub-strategies. Instead, they adopted a hierarchical reinforcement learning (HRL) approach based on Multi-level Control to abstract the decision-making process into two control levels, with the upper-level strategy controlling the lower-level strategy.

In order to address this issue, Liao et al. [31] simultaneously trained both levels of strategy. The high-level strategy is referred to as the main controller, responsible for triggering sub-controllers in the lower level. The lower-level sub-controllers comprise N symptom checkers and one disease classifier. When the main controller deems that enough symptom information has been collected, it triggers the lower-level disease classifier to make the final diagnosis. Otherwise, it triggers the symptom checkers to continue asking about symptoms. The introduction of hierarchical reinforcement learning has significantly improved the diagnostic accuracy of intelligent diagnostic dialogue systems when dealing with a large number of diseases and symptoms.

Another related area is the usage of knowledge graphs in diagnostic systems [22,34,35]. Xu et al. [22] proposed a knowledge-guided KR-DS dialogue system, which combined a knowledge graph with the dialogue system to improve the diagnostic accuracy of intelligent diagnostic dialogue systems. Their KR-DQN model extended the single DQN agent by incorporating a medical knowledge graph. The graph includes two types of nodes: diseases and symptoms, both of which are connected by edges representing symptom–symptom and disease–symptom relationships. To capture the correlation between different symptoms and diseases, KR-DS calculated conditional probabilities as edge weights based on the information in the dataset.

Zhao et al. [36,37] argued that KR-DS only utilized the statistical features of the dataset to construct the knowledge graph, which lacks an in-depth understanding of disease–symptom and disease–disease relationships. To address this limitation, they applied Graph Convolutional Networks (GCNs) [38–40] to learn node representations in the knowledge graph. They combined GCN with the DQN model, proposing the Graph–DQN model.

Liu et al. [41] also introduced a knowledge graph in their model, considering two types of logic in symptom inquiry. One type is to ask for confirmation of a specific disease, which can be represented using conditional probabilities such as in KR-DS [22]. The other type is to inquire about distinguishing several similar diseases, which can be represented using mutual information between diseases and symptoms. The correlation between these two types of diseases and symptoms is combined in a weighted manner, forming the weights in the knowledge graph.

To enhance the acceptability of the medical system, Arsène Fansi Tchango et al. [42,43] incorporated the doctor's trust reasoning as a requirement by introducing a reinforcement learning model and redefining the evidence gathering and automated diagnosis tasks. They trained an agent to improve the system's performance through predicting discrepancies and shaping rewards. However, due to their overly strict reward function setting, they failed to address the sparsity issue of rewards, which affected the inference performance of the model.

Our model combines the aforementioned hierarchical reinforcement learning with action space pruning, effectively addressing the curse of dimensionality in both action and

state spaces. To tackle the reward sparsity issue in the disease classifier, we propose an entropy-based reward function that ensures progressive convergence towards accurate disease classification during patient interactions, thus ensuring algorithmic correctness. Finally, we validate the effectiveness of our algorithm on the SD dataset.

3. Methodology

3.1. Overview

The architecture of the hierarchical reinforcement-learning-based intelligent diagnostic dialogue algorithm is shown in Figure 1, consisting of four main components: the agent, disease classifier, state tracker, and patient simulator. The agent and disease classifier together form the dialogue policy component, the state tracker is responsible for maintaining and updating the current environment state, and the patient simulator generates patient actions and provides feedback to the agent as rewards.

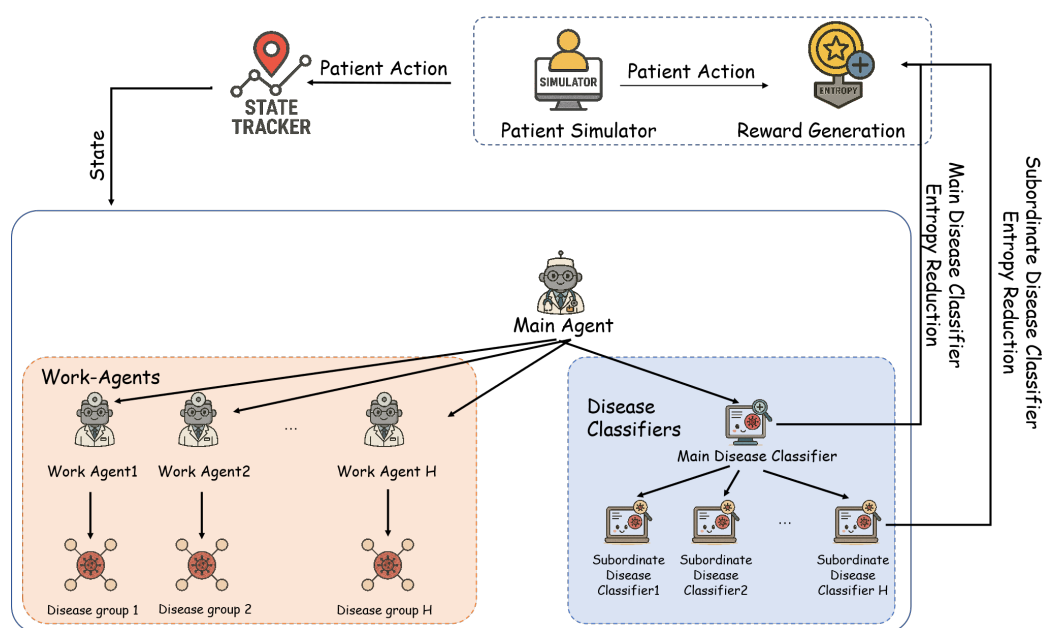


Figure 1. Architecture of MA-HRL.

First, the dialogue policy component is treated as a complete agent. The agent receives the current environment state at time step t from the state tracker and maps the state to an action $a_t \in A$ based on the policy $\pi(a | s_t)$. The patient simulator generates patient actions and returns the computed reward r_t to the agent, after which the environment state transitions to $s_t \in S$. The goal of the agent is to find an optimal policy that maximizes the expected cumulative reward $r_t = R(s_t, a_t)$, where $\gamma \in [0, 1]$ is the discount factor and T is the maximum number of dialogue turns. In the context of intelligent diagnosis dialogue, the environment state mainly includes the symptoms reported by the patient that the dialogue system has obtained. Let D be the set of diseases and SY be the set of symptoms associated with these diseases. The state $s_t = [b_1, b_2, \dots, b_{|SY|}, t, re]$ is represented as follows: b_i is a one-hot vector of length 3 ($i = 1, 2, \dots, |SY|$), where $b_i \in \{[1, 0, 0], [0, 1, 0], [0, 0, 1], [0, 0, 0]\}$ indicates the presence of a certain condition, as shown in Table 1. In addition, the state vector includes the current turn t and whether there was a repetition of questions in the previous turn $re \in \{0, 1\}$.

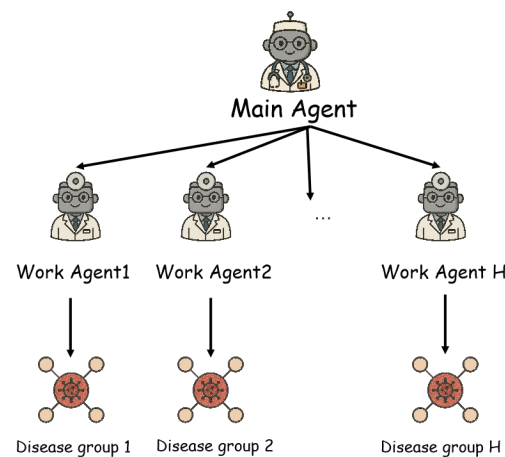
Table 1. Meaning of state vector.

Vector	Meaning
$[0, 0, 0]$	No inquiry about this symptom
$[0, 0, 1]$	Patient responds “Uncertain”
$[0, 1, 0]$	Patient responds “No”
$[1, 0, 0]$	Patient responds “Yes”

The action space $A = D \cup SY$ of the agent consists of two types of actions: if $a_t \in SY$, the dialogue system selects a symptom to ask the patient about; if $a_t \in D$, the dialogue system makes a final diagnosis, and the dialogue process ends. The success or failure of the entire dialogue process is determined by whether the diagnosis result is correct.

3.2. Hierarchical Reinforcement Learning Architecture

In this study, the disease set D is divided into H groups, where $D = D_1 \cup D_2 \cup \dots \cup D_H$, and $D_i \cap D_j = \emptyset, i \neq j, i = 1, 2, \dots, H, j = 1, 2, \dots, H$. Each sub-disease set D_i is associated with a group of symptoms. The hierarchical architecture adopts a two-level structure, with H Worker Agent in the lower level, each corresponding to a disease group. Each Worker Agent is only concerned with the diseases in its corresponding group and the symptoms associated with those diseases, as shown in Figure 2.

**Figure 2.** Hierarchical reinforcement learning architecture.

In each round, the primary agent receives the environment state s_t , which represents all the currently obtained symptom information. Based on the policy $\pi^M(a_t^M | s_t)$, the primary agent decides whether to continue asking symptoms or perform disease diagnosis. The action space of the primary agent is defined as shown in Equation (1), which includes all the Worker Agents and the disease classifier. If one of the Worker Agents is triggered, a subtask is initiated, and in the following k rounds, the symptom inquiry will be conducted by the same Worker Agent ($k \in [1, N]$, where N represents the maximum number of rounds for the subtask). The current subtask ends when the response to the symptom inquiry by the Worker Agent is “yes” or “no”. If the disease classifier is triggered, the dialogue system provides a disease judgment, and the current dialogue ends.

$$A^M = \{W_1, W_2, \dots, W_H, \text{disease_classifier}\} \quad (1)$$

When a Worker Agent is triggered, it receives the environment state s_t and extracts the relevant symptoms s_t^i corresponding to its assigned subgroup ($s_t^i = [b_1, b_2, \dots, b_{|SY_i|, t, re}]$). The Worker Agent selects a symptom from its own action space based on the policy

$\pi^i(a_t^i | s_t^i)$ for further inquiry. The action space of the Worker Agent is defined as shown in Equation (2).

$$A^i = \{sy_1, sy_2, \dots, sy_{|SY_i|}\}, i = 1, 2, \dots, H \quad (2)$$

In the hierarchical reinforcement learning architecture, the main agent at the higher level receives an external reward r_t^e at each round. After the completion of a sub-task, the main agent calculates the cumulative reward r_t^M for that round using Equation (3), where $i = 1, 2, \dots, H$, and γ is the discount factor.

$$r_t^M = \begin{cases} \sum_{t'=1}^k \gamma^{t'} r_{t+t'}^e, & \text{if } a_t^M = W_i \\ r_t^e, & \text{if } a_t^M = \text{disease_classifier} \end{cases} \quad (3)$$

The objective of the main agent is to maximize its reward; therefore, its loss function is computed using Equation (4), where s' represents the next dialogue state, $a^{M'}$ is the next action of the main agent, $\theta_{\bar{M}}$ denotes the parameters of the target network, θ_M represents the parameters of the current network, and $\mathcal{B}^M$ refers to the replay buffer of the main agent.

$$\mathcal{L}(\theta_m) = \mathbb{E}_{s, a^M, r^M, s' \sim \mathcal{B}^M} \left[\left(r^M + r_M^N \max_{a^M} Q_m(s', a^{M'}; \theta_m') - Q_m(s', a^M; \theta_m) \right)^2 \right] \quad (4)$$

The low-level Worker Agents receive internal rewards r_t^i at each round. The objective of the Worker Agents is to maximize their rewards, and their loss function is computed using Equation (5). In this equation, θ^{i-} represents the parameters of the target network, θ^i represents the parameters of the current network, and $\mathcal{B}^i$ refers to the replay buffer of the Worker Agents.

$$\mathcal{L}(\theta_i) = \mathbb{E}_{s^i, a^i, r^i, s^{i'} \sim \mathcal{B}^i} \left[\left(r^i + r_W \max_{a^{i'}} Q_i(s^{i'}, a^{i'}; \theta^{i-}) - Q_i(s^i, a^i; \theta^i) \right)^2 \right] \quad (5)$$

3.3. Hierarchical Disease Classifier

If a single agent is used in the dialogue management module, the agent's action space includes both symptoms and diseases. However, by assigning the disease classifier the task of disease diagnosis, the action space of the agent is narrowed, alleviating the curse of dimensionality issue. Unlike the agent, the disease classifier operates differently. While the agent optimizes its strategy by maximizing rewards within the reinforcement learning framework, disease classification involves supervised learning, where collected symptom information serves as inputs to the classifier. The classifier's output is a probability distribution over all diseases, and the dialogue management module selects the disease with the highest probability as the final diagnosis.

Inspired by hierarchical reinforcement learning, this study applies hierarchical thinking to the process of disease diagnosis, modeling disease classifiers in a hierarchical manner, as depicted in Figure 3. The higher-level primary classifier, with a single instance, yields a probability distribution vector $p_m = [p_1, p_2, \dots, p_H]$ over H disease groups. Among these, the group with the highest value triggers its corresponding secondary disease classifier, producing a probability distribution vector $p_i = [p_1, p_2, \dots, p_{|D_i|}]$ for diseases within that group. The loss function encompasses the categorical cross-entropy between the output probability distribution and the actual distribution. The cross-entropy for the primary disease classifier is expressed as in Equation (6), while for the secondary disease classifier, it follows Equation (7). Here, $y_j \in \{0, 1\}$. For the primary disease classifier, if the true group

of the disease is j , y_j takes the value 1; otherwise, it takes 0. For the secondary disease classifier, y_j is 1 if the disease equals j ; otherwise, it is 0.

$$\mathcal{L}_m = - \sum_{j=1}^H y_j \log(p_j) \quad (6)$$

$$\mathcal{L}_i = - \sum_{j=1}^{|D_i|} y_j \log(p_j) \quad (7)$$

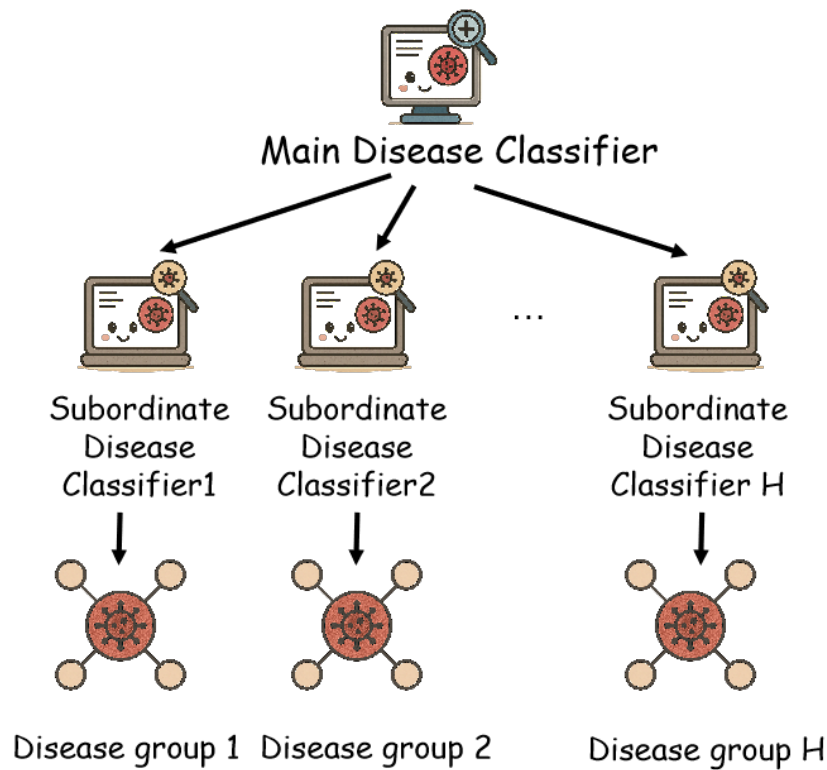


Figure 3. Hierarchical diagnostic algorithm.

3.4. Reward Design Based on Information Entropy Differential

In the hierarchical architecture, based on the initial state, the initial probability distributions computed by the main disease classifier and the auxiliary disease classifiers are denoted as $p_{M0} = [p_1, p_2, \dots, p_H]$ and $p_{W0} = [p_1, p_2, \dots, p_{|D_i|}]$, respectively. Their entropies can be calculated using Equations (8) and (9), respectively. The calculated initial entropies serve as normalization terms between different dialogues, preventing significant discrepancies in reward values across different dialogues [44].

$$H_M(s_0) = - \sum_{i=1}^H p_i \log(p_i) \quad (8)$$

$$H_W(s_0) = - \sum_{i=1}^{|D_i|} p_i \log(p_i) \quad (9)$$

Similarly, at time t , the information entropy of the main disease classifier $H_M(s_t)$ and the auxiliary disease classifier $H_W(s_t)$ are computed. After the dialogue system takes an action and transitions to a new state s_{t+1} , the information entropies $H_M(s_{t+1})$ and $H_W(s_{t+1})$ are calculated. The information discrepancy rewards for the main agent and the auxiliary agent are then obtained using Equations (10) and (11), respectively. The term

$H_{\Phi}(s_t) - H_{\Phi}(s_{t+1})$ calculates the difference in entropy between two consecutive time steps. Ideally, this difference should always be positive, but in practice, there may be cases where the entropy increases. Therefore, the max function is used to avoid negative rewards.

$$r_{entropy}^M = \max\left(\frac{H_M(s_t) - H_M(s_{t+1})}{H_M(s_0)}, 0\right) \quad (10)$$

$$r_{entropy}^W = \max\left(\frac{H_W(s_t) - H_W(s_{t+1})}{H_W(s_0)}, 0\right) \quad (11)$$

In summary, the external reward function is designed in this study as shown in Equation (12), and the internal reward function is formulated as shown in Equation (13), where N represents the maximum number of dialogue turns:

$$r_t^e = \begin{cases} +3 \times N + r_{entropy}^M & \text{if success} \\ -2 \times N + r_{entropy}^M & \text{if repeated} \\ -3 \times N + r_{entropy}^M & \text{if reach max turn} \\ r_{entropy}^M & \text{otherwise} \end{cases} \quad (12)$$

$$r_t^{in} = \begin{cases} +2 \times N + r_{entropy}^W & \text{if match} \\ -2 \times N + r_{entropy}^W & \text{if repeated} \\ r_{entropy}^W & \text{otherwise} \end{cases} \quad (13)$$

In addition to the information entropy difference reward, the main agent is designed with positive rewards for successful diagnosis and negative rewards for diagnosis failure or exceeding the maximum number of dialogue turns. For the Worker Agents, positive rewards are assigned for inquiring about symptoms with patient responses of “yes” or “no,” while negative rewards are assigned for repeating previously asked symptoms.

3.5. Design of Knowledge Graph for Dialogue Management Module

Medical knowledge can be represented in the form of a knowledge graph, which can be utilized by the dialogue system. In this study, the diseases and symptoms extracted from the training and testing datasets are treated as nodes in the knowledge graph, and the related diseases and symptoms are connected through edges. To characterize the correlation between different symptoms and diseases, weights are assigned to the edges as edge attributes. Since a hierarchical structure is employed in this study, the Worker Agents at the lower level are responsible for a subset of diseases and symptoms. Therefore, a separate medical knowledge graph is constructed for each Worker Agent, which only includes the disease set D_i and symptom set SY_i associated with that Worker Agent. The weights of the edges represent the degree of correlation between symptoms and diseases. In this study, the conditional probability $p(sy_i | d_j)$ is computed based on the dialogue dataset and used as the edge weight. $p(sy_i | d_j)$ indicates the probability of a patient having symptom sy_i given that they have disease d_j , and its calculation is described by Formula (14). Here, $n(d_j, sy_i)$ represents the number of dialogues where the patient has disease d_j and exhibits symptom sy_i . By calculating the probability for each symptom–disease pair, the conditional probability matrix $M_i \in \mathbb{R}^{|SY_i| \times |D_i|}, i = 1, 2, \dots, H$ for the disease group is obtained.

$$p(sy_i | d_j) = \frac{n(d_j, sy_i)}{\sum_{k=1}^N n(d_j, sy_k)} \quad (14)$$

Based on Formula (15), the current symptom probability distribution can be calculated, where $p(dis)$ is the probability distribution computed by the corresponding sub-disease classifier $p(dis) \in \mathbb{R}^{|D_i| \times 1}$, $p(sym) \in \mathbb{R}^{|SY_i| \times 1}$.

$$p(sym) = M_i \cdot p(dis) \quad (15)$$

The probability distribution $p(sym)$ is combined with the Q-values computed by the Worker Agents, using prior knowledge to assist the agent's decision-making. The calculation formula is shown in Equation (16). $MLP(s_t)$ represents the Q-value distribution computed by the Worker Agents, and the maximum value is selected to correspond to the symptom for inquiry.

$$a_t = \operatorname{argmax}_a (\operatorname{sigmoid}(MLP_W(s_t)) + \operatorname{sigmoid}(p(sym))) \quad (16)$$

4. Experimental

4.1. Dataset

The SD dataset [31] has been used to test the model. The SD dataset is generated from the medical database SymCat (www.symcat.com) including 30,000 dialogue data points involving 90 diseases and 266 symptoms. The construction process of the dataset is as follows: In the SymCat database [45], there are 801 diseases, which were divided into 21 groups based on the International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM) [46]. The most common 9 groups were selected, and 10 most common diseases were chosen from each group, resulting in a total of 90 diseases. In the Centers for Disease Control and Prevention (CDC) database [47], each disease is associated with a set of symptoms, and each symptom is assigned a probability value indicating the likelihood of having that symptom when diagnosed with the disease. Given a disease and its associated symptoms, the generation process of a patient's goal is as follows: For each symptom, a random true or false label is assigned (indicating the presence or absence of the symptom), and then one symptom is randomly selected from all the symptoms marked as true as the explicit symptom, while the others are considered implicit symptoms.

Table 2 shows an overview of the dataset, and Table 3 shows the number of symptoms in each disease group, including the number of unique symptoms, which refers to the symptoms not related to other disease groups. It can be observed that each group has relatively fewer unique symptoms compared to associated symptoms. This also increases the difficulty for the intelligent conversational system to accurately identify the correct disease as a symptom is often associated with multiple disease groups.

Table 2. Overview of the dataset.

Categories	Quantity
Patient goal	30,000
Disease	90
Disease group	10
Symptoms	266
Average prominent symptoms	2.6

Table 3. Distribution of disease groups in the dataset.

Disease Groups	Disease Group Names	Number of Disease Categories	Number of Patient Goals	Number of Associated Symptoms	Number of Unique Symptoms
1	Infectious diseases and parasitic diseases	10	3371	65	9
4	Endocrine, nutritional and metabolic diseases	10	3348	89	16
5	Psychiatric and behavioral disorders	10	3355	68	10
6	Neurological disorders	10	3380	58	7
7	Ocular and adnexal diseases	10	3286	46	10
12	Skin and subcutaneous tissue diseases	10	3303	51	18
13	Musculoskeletal system and connective tissue diseases	10	3249	62	24
14	Diseases of the genitourinary system	10	3274	69	26
19	Injuries, poisonings, and certain other external causes	10	3389	73	6

4.2. Experimental Configuration

We use 80% of the data from the SD dataset, which include 24,000 patient goals, for training purposes. The remaining 20% of the data, amounting to 6000 patient goals, is used for testing. A total of 5000 training episodes are conducted, with each episode randomly selecting 100 patient goals for 100 independent dialogues. In order to ensure sufficient exploration and avoid local optima, both the primary agent and the secondary agent utilize the ϵ -greedy strategy during training. This means that the agents randomly select actions with a probability of ϵ and act according to the current policy with a probability of $1 - \epsilon$.

Other hyperparameters are shown in Table 4.

Table 4. Model hyperparameter settings.

Hyperparameters	Values
Training epochs	5000
Number of dialogues per epoch	100
Batch size	100
Learning rate	0.005
ϵ	0.1
Maximum dialogue turns	28
Maximum dialogue turns for subtask N	5
Discount factor γ	0.95

4.3. Indicators

To evaluate the performance of the dialogue system and compare it with other models, certain metrics need to be defined. In each episode, K independent dialogues are conducted. This study focuses on the following metrics for assessment:

1. The success rate of the dialogue system, specifically the diagnostic accuracy in the intelligent medical inquiry task, is considered as one of the evaluation metrics.

$$\frac{\sum_{i=1}^K success_i}{K} \times 100\% \quad (17)$$

where $success_i = 1$ if the dialogue system performs a successful diagnosis; otherwise, $success_i = 0$.

2. The cost of dialogue generation, specifically the number of dialogue turns.

3. Symptom hit rate, which refers to the ratio of symptoms inquired by the dialogue system for which the patient actually responds “yes”.

$$\frac{\sum_{i=1}^K \frac{m_i}{r_i}}{K} \times 100\% \quad (18)$$

where m_i represents the number of symptoms in the i th dialogue for which the patient responds “yes,” and r_i represents the total number of inquired symptoms.

4. Symptom recognition rate refers to the ratio of symptoms for which the patient responds “yes” to the total number of implicit symptoms that the patient has.

$$\frac{\sum_{i=1}^K \frac{d_i}{r_i}}{K} \times 100\% \quad (19)$$

where d_i represents the number of symptoms for which the patient responds “yes” in the i th dialogue, and r_i represents the total number of implicit symptoms the patient has.

Among them, the success rate of the dialogue system, which refers to the accuracy of the diagnosis, is a key indicator that determines the performance of the model in intelligent medical consultation tasks. It directly affects the usability of the intelligent medical consultation dialogue system.

4.4. Performance Analysis

The comparative experiment compares the MA-HRL model in this study with the following four related models:

- SVM-ex: This model uses the Support Vector Machine (SVM) model [48] and treats the intelligent medical consultation task as a supervised classification task. The input consists of the patient’s explicit symptoms, and the output is the disease diagnosis. The experimental results of this model are generally considered as the lower bound of the diagnostic accuracy in intelligent medical consultation tasks.
- SVM-ex&im: This model is similar to the SVM-ex model, with the difference that the input includes both explicit and implicit symptoms. The experimental results of this model are generally considered as the upper bound of the diagnostic accuracy in intelligent medical consultation tasks.
- Flat-DQN [21]: This model uses a single agent to make decisions, and the action space consists of a set of diseases and symptoms. Its neural network structure is the same as the agent in this study.
- HRL [31]: HRL is currently the state-of-the-art method, which first proposed the SD dataset and trained and validated the model on this dataset. HRL adopts a hierarchical reinforcement learning architecture, and its main and Worker Agents have the same neural network structure as in this study. However, its disease classifier is not hierarchical but a single neural network with a three-layer structure and 256 neurons in the hidden layer. HRL does not introduce information entropy and does not include any knowledge graph.

As shown in Table 5, the experimental results demonstrate that the MA-HRL model in this study outperforms the HRL model and the Flat-DQN model in all other metrics except the number of dialogue turns. Although there is a slight increase in the number of dialogue turns compared to HRL, it only incurs an average of one additional dialogue turn. On the other hand, while the Flat-DQN model has fewer dialogue turns, its diagnostic accuracy is lower. Compared to the HRL model, the MA-HRL model achieves a 7.2% improvement in diagnostic accuracy, a 0.91% improvement in symptom hit rate, and a significant 15.94% improvement in symptom recognition rate. The significant improvement in symptom recognition rate indicates that the MA-HRL model can inquire about nearly half of the patients’ implicit symptoms. Figure 4 presents the comparative experimental results of the three reinforcement learning models in the form of a bar chart.

Table 5. Results of comparative experiments.

Model	Diagnostic Accuracy (%)	Number of Dialogue Turns	Symptom Matching Rate (%)	Symptom Recognition Rate (%)
SVM-ex	32.2	N/A	N/A	N/A
SVM-ex&im	73.1	N/A	N/A	N/A
Flat-DQN	37.7	6.2	3.1	5.6
HRL	49.5	12.95	10.49	29.56
MA-HRL (Ours)	56.7	13.55	11.4	45.5

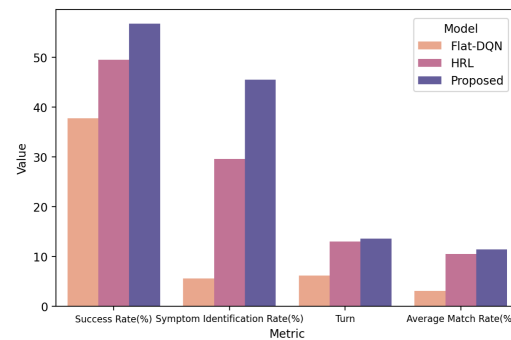
**Figure 4.** Comparison experimental results.

Figure 5 depicts the learning curve comparisons of the MA-HRL model, Flat-DQN, and HRL in this study.

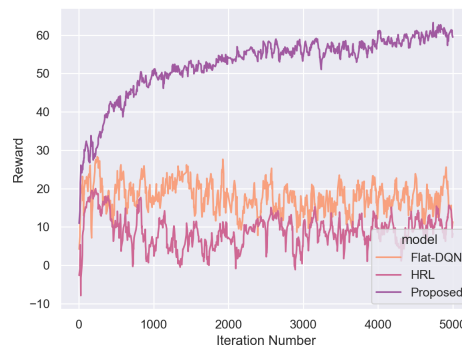
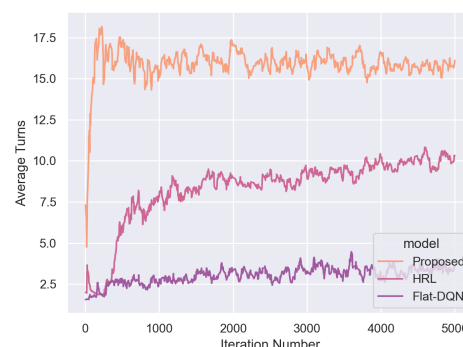
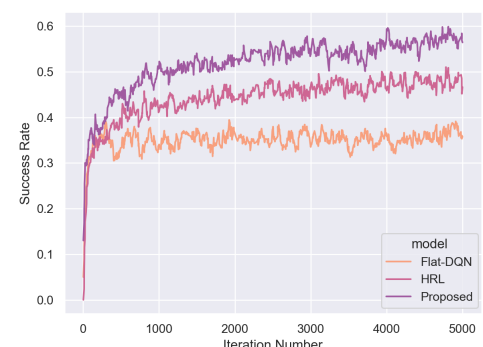
**(a)** Reward changes during model learning**(b)** Dialogue turn shifts during training**(c)** Model diagnostic accuracy changes**Figure 5.** Learning curve comparisons of the MA-HRL, Flat-DQN, and HRL.

Figure 5 shows the variation of the model's average reward values with the number of training rounds. Our proposed model obtained substantial rewards early in the training process, resulting in a significant improvement compared to the baseline. Figure 5b

illustrates the change in the average number of dialogue turns with the number of training rounds, demonstrating a notable improvement in our model as well. In Figure 5c, the success rate of disease diagnosis is depicted as it changes with the number of training rounds. Our model achieved a 10% increase in accuracy compared to the baseline. In summary, our model exhibits substantial improvements in terms of reward values, dialogue turns, and accuracy.

4.5. Ablation Experiment

The ablation experiments aim to compare and validate the contributions of multiple improvements made to the experimental metrics. The ablation experiments will compare the following models, where 1, 2, and 3 each add one improvement to the HRL model and 4, 5, and 6 add two improvements to the HRL model. Model 7 represents the complete model proposed in this study:

1. HRL + KG: Adding medical knowledge graph to the HRL model. 2. HRL + reward: Adding information gain reward to the HRL model. 3. HDC: Adding hierarchical disease classifier to the HRL model. 4. HRL + KG + reward: Adding medical knowledge graph and information gain reward to the HRL model. 5. HDC + KG: Adding medical knowledge graph to the HDC model. 6. HDC + reward: Adding information gain reward to the HDC model. 7. MA-HRL (Ours): The complete model proposed in this study.

The experimental results, as shown in Table 6, reveal that when only one improvement is added to the HRL model, the introduction of medical knowledge graph contributes significantly to the enhancement of the metrics, with higher symptom hit rate and symptom recognition rate compared to the other two models. However, in the comparison of models with two improvements, the model incorporating the hierarchical disease classifier performs better, while the model with information gain reward outperforms the model with knowledge graph in all metrics. Figure 6 presents the results of the ablation experiments in a bar chart format.

Table 6. Ablation Study Results.

Model	Diagnostic Accuracy (%)	Number of Dialogue Turns	Symptom Matching Rate (%)	Symptom Recognition Rate (%)
HRL + KG	55.0	12.64	12.0	45.0
HRL + reward	54.4	12.62	11.2	42.6
HDC	53.9	12.94	10.7	42.9
HRL + KG + reward	54.6	13.20	10.7	41.4
HDC + KG	55.2	13.84	10.4	44.8
HDC + reward	55.3	13.23	12.0	45.5
MA-HRL (Ours)	56.7	13.55	11.4	45.5

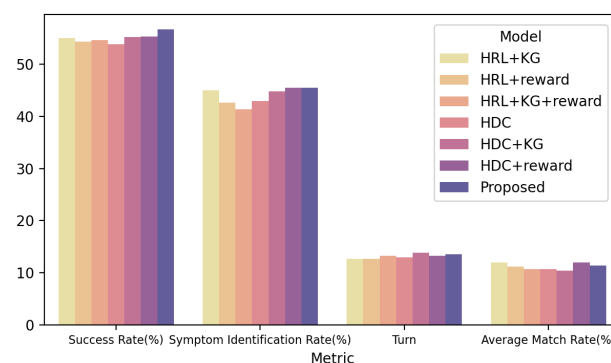


Figure 6. Results of ablation experiments.

Table 7 presents the models that include a hierarchical disease classifier, along with the performance of the main disease classifier and each subordinate disease classifier during testing. It can be observed that the disease classifiers of each model exhibit similar performance across different disease groups. For example, they perform best in disease group 4 (endocrine, nutritional, and metabolic diseases) and worst in disease group 7 (eye and adnexa diseases). The main classifier and the three subordinate classifiers of the proposed model in this study achieve higher accuracy compared to other models, possibly due to the intelligent agent obtaining more symptom information through questioning. Overall, the performance of the disease classifiers in other models can be ranked from best to worst as HDC + KG, HDC, and HDC + reward, which aligns with the diagnostic accuracy of each model. Figure 7 visualizes these experimental results in a bar chart format.

Table 7. Accuracy of each disease classifier.

Disease Classifier	HDC	HDC + KG	HDC + Reward	MA-HRL (Ours)
Primary disease classifier	75.3%	77.1%	76.1%	77.9%
Secondary disease classifier1	69.94%	70.68%	70.24%	78.42%
Secondary disease classifier 4	91.3%	92.04%	92.48%	94.4%
Secondary disease classifier 5	62.36%	63.3%	63.9%	66.42%
Secondary disease classifier 6	80.65%	79.77%	77.27%	79.91%
Secondary disease classifier 7	55.75%	55.46%	60.47%	58.85%
Secondary disease classifier 12	63.6%	65.38%	62.85%	64.64%
Secondary disease classifier 13	78.85%	76.67%	77.14%	74.96%
Secondary disease classifier 14	85.92%	86.39%	85.6%	85.6%
Secondary disease classifier 19	76.53%	78.33%	74.59%	77.88%

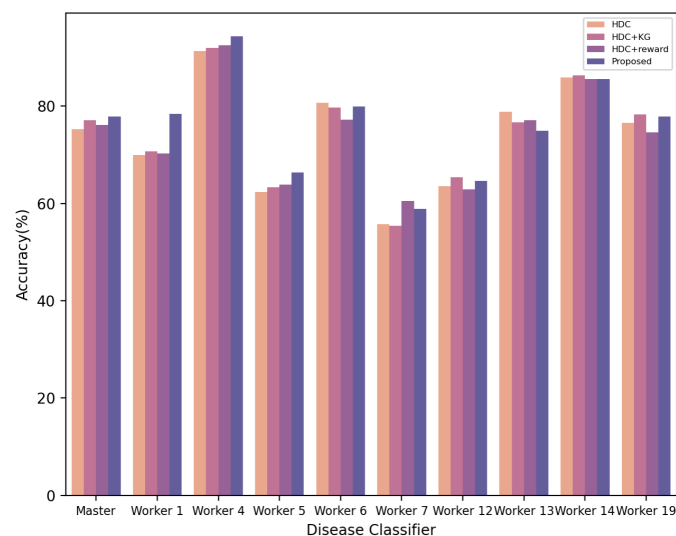


Figure 7. Comparison of accuracy among different disease classifiers.

To further investigate the performance of the disease classifiers in the proposed model, the alignment between the actual disease group of the selected patient targets during testing and the predicted disease group by the disease classifiers is presented in Figure 8. From the figure, it can be observed that, due to the 77.9% accuracy of the main disease classifier, in most cases, the predicted disease group aligns with the actual disease group. However, inaccuracies in prediction occur due to the lower accuracy of some subordinate classifiers.

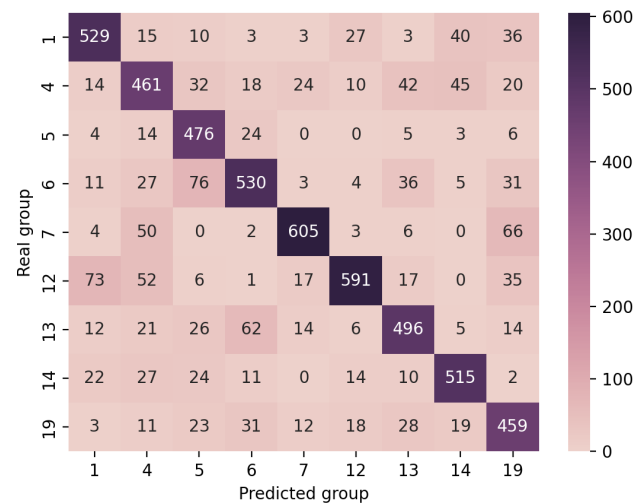


Figure 8. Prediction of disease group by disease classifiers.

5. Conclusions

This study proposes and implements an intelligent diagnosis dialogue algorithm based on deep reinforcement learning. Inspired by hierarchical reinforcement learning, a hierarchical architecture for intelligent agents and disease classifiers is designed. The algorithm introduces the concept of information entropy and designs a reward function based on the difference in information entropy. A medical knowledge graph is constructed to assist intelligent agents in decision-making. The algorithm achieves effective results on a real-world dataset. However, current intelligent diagnosis systems typically only consider the patient's symptom information as input and overlook other patient-specific features such as age, gender, and pathography. Incorporating these additional information into the system to improve the diagnostic accuracy is one of the future directions for further research.

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